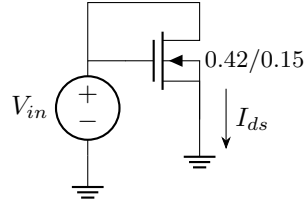


EC 5311 Digital IC Design: Assignment 1
Transfer characteristics of transistors

1. (a) Consider the nMOS transistor connected as shown below.



Is it in the saturation or linear region? Using ngspice, obtain the transfer characteristics I_{ds} vs V_{gs} assuming $L = 0.15\mu\text{m}$ and $W = 0.42\mu\text{m}$. Plot the simulated model.